Finance Concepts for Banking Analytics: Interview Guide

Formulas, worked examples and real-world uses

How to use this guide. Every topic follows the same layout: **Idea → Formula → Worked example → Real-world use → Interview questions.** The numbers match the notebook `Financial_Modeling_Toolkit.ipynb`, so you can say "I built this and got X."

1. Time Value of Money

Idea. A dollar today is worth more than a dollar tomorrow, because today's dollar can earn interest and tomorrow's is exposed to inflation and risk. Discounting converts future money into today's money.

Formulas

Future value and present value:

$$FV = PV\,(1+r)^{n} \qquad\qquad PV = \frac{FV}{(1+r)^{n}}$$

Annuity (equal payment C each period for n periods):

$$PV = C \cdot \frac{1-(1+r)^{-n}}{r}$$

Perpetuity and growing perpetuity (requires $r > g$):

$$PV = \frac{C}{r} \qquad\qquad PV = \frac{C_1}{r-g}$$

Effective annual rate from a quoted rate (APR) compounded m times a year:

$$EAR = \left(1+\frac{APR}{m}\right)^{m} - 1$$

Worked example

Question	Calculation	Answer
$10,000 at 5% for 10 years	$10{,}000 \times 1.05^{10}$	**$16,288.95**
$1,000/yr for 20 years at 7%	$1{,}000 \times \frac{1 - 1.07^{-20}}{0.07}$	**$10,594.01**
6% APR compounded monthly	$(1.005)^{12} - 1$	**6.17% EAR**

Real-world use

- **Savings and loan products.** A bank advertises "6% compounded monthly." The customer really earns 6.17%. Regulators require banks to show APR and APY so customers can compare products honestly.
- **Pricing a bond or a pension.** A bond's price is the PV of its coupons (an annuity) plus its face value. A pension fund discounts promised future payouts to see whether it has enough assets today.
- **Stock valuation.** The dividend discount model treats a stock as a growing perpetuity: price $= D_1/(r - g)$.

Interview questions. Why does a higher discount rate lower PV? What is the difference between APR and APY? How is a bond an annuity plus a lump sum?

2. Loan Amortization

Idea. A fixed-payment loan charges interest on the *remaining balance*, so early payments are mostly interest and later ones mostly principal.

Formulas

Level monthly payment on loan L, with monthly rate $r = APR/12$ and n months:

$$P = L \cdot \frac{r}{1 - (1 + r)^{-n}}$$

Month-by-month split:

$$\text{interest}_t = r \cdot B_{t-1} \qquad \text{principal}_t = P - \text{interest}_t \qquad B_t = B_{t-1} - \text{principal}_t$$

Worked example. Loan $L = \$400{,}000$, 6.5% APR, 30 years ($r = 0.005417$, $n = 360$):

$$P = 400{,}000 \cdot \frac{0.005417}{1 - (1.005417)^{-360}} = \$2{,}528.27$$

Month 1: interest $= 400{,}000 \times 0.005417 = \$2{,}166.67$, so principal is only **$361.60**, about 14% of the payment. Total interest over 30 years is **$510,178**, more than the loan. An extra $300/month shortens the loan by **7.6 years** and saves about **$149,581**.

Real-world use

- **Mortgage and auto-loan desks** generate customer schedules with exactly this formula.
- **Prepayment risk.** When rates fall, borrowers refinance and repay early. The bank loses expected interest income and must reinvest at lower rates. Mortgage-backed securities are priced on prepayment assumptions for this reason.
- **Customer advice.** Showing a customer that a small extra payment saves six figures is a common retention and financial-wellness message.

Interview questions. Why is early payment mostly interest? Fixed versus variable rate: who bears the interest-rate risk? (With fixed rates, the bank does, unless it hedges.) How does prepayment change an asset's duration?

3. Capital Budgeting: NPV, IRR, Payback

Idea. Does a project's stream of discounted cash flows exceed what it costs? Banks decide on branches, technology and acquisitions this way.

Formulas

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}$$

IRR is the rate that makes NPV zero:

$$0 = \sum_{t=0}^{T} \frac{CF_t}{(1+IRR)^t}$$

$$\text{Profitability index} = \frac{PV(\text{inflows})}{\text{initial cost}}$$

Decision rule: accept if $NPV > 0$, or equivalently if $IRR >$ the hurdle rate (usually WACC).

Worked example. Cash flows $-1000,\ 400,\ 500,\ 600$ at $r = 10\%$:

$$NPV = -1000 + \frac{400}{1.1} + \frac{500}{1.1^2} + \frac{600}{1.1^3} = -1000 + 363.64 + 413.22 + 450.79 = \$227.65$$

Solving for the zero gives **IRR = 21.65%**, well above the 10% hurdle, so accept.

Real-world use

- **Technology investment.** A bank compares a $2.0M digital-banking platform (cash flows rising to $1.0M a year) with a $1.5M branch expansion. At a 9% WACC, the platform has NPV $1.0M versus $0.4M for the branch (see the notebook), so it adds more value.
- **Corporate lending.** Credit analysts compute NPV and IRR of a borrower's project to judge whether it can service the loan.

Interview questions. Why prefer NPV over IRR? (NPV measures dollars of value created; IRR can give multiple answers when cash flows change sign more than once, and can mis-rank projects of different size.) How do you compare projects with unequal lives? (Equivalent annual annuity.)

4. Bonds, Duration and Convexity

Idea. A bond pays coupons then principal. When market yields rise, existing bonds lose value. Banks hold large bond portfolios, so this is a core interest-rate risk.

Formulas

Price at yield y, with m payments a year and T years:

$$P = \sum_{k=1}^{mT} \frac{C/m}{(1+y/m)^{k}} + \frac{F}{(1+y/m)^{mT}}$$

Macaulay duration (PV-weighted average time to receive the cash, in years), and modified duration:

$$D_{Mac} = \frac{1}{P}\sum_{t} t \cdot \frac{CF_t}{(1+y/m)^{mt}} \qquad D_{mod} = \frac{D_{Mac}}{1+y/m}$$

Price change for a yield change Δy:

$$\frac{\Delta P}{P} \approx -D_{mod}\,\Delta y + \tfrac{1}{2}\,\text{Convexity}\,(\Delta y)^2$$

DV01 (dollar change per one basis point): $DV01 = P \cdot D_{mod} \cdot 0.0001$.

Worked example. 10-year, 5% coupon, semi-annual, yield 5%: price $1,000, $D_{Mac} = 7.99$ years, $D_{mod} = 7.80$, convexity $= 73.6$.

$$\Delta y = +1\%: \quad -7.80\% + \tfrac{1}{2}(73.6)(0.01)^2 = -7.43\% \quad (\text{exact: } -7.44\%)$$

$$\Delta y = -1\%: \quad +7.80\% + 0.37\% = +8.17\% \quad (\text{exact: } +8.18\%)$$

Convexity helps in both directions: prices rise more than they fall.

Real-world use

- **Treasury and ALM.** A bank with $5B of bonds at $D_{mod} = 5$ loses about $250M if yields jump 100bp. The ALM committee limits this with hedges (interest-rate swaps) or by shortening the portfolio.
- **2022–2023 bank failures.** Silicon Valley Bank held long-duration bonds funded by short-term deposits. Rates rose, bond values fell, deposits fled, and losses were realised. This is the standard case study for duration risk.
- **Trading desks** quote risk in DV01, not in price.

Interview questions. Why are long, low-coupon bonds riskier? What is the duration of a zero-coupon bond? (Its maturity.) How would you hedge a duration mismatch? What happened at SVB?

5. Risk, Return and CAPM

Idea. Investors are paid only for risk they cannot diversify away (market risk). Beta measures how much a stock moves with the market.

Formulas

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} \qquad E[R_i] = r_f + \beta_i\,(E[R_m] - r_f)$$

$$\text{Sharpe} = \frac{R_p - r_f}{\sigma_p} \qquad \text{Treynor} = \frac{R_p - r_f}{\beta_p} \qquad \alpha_{Jensen} = R_p - [r_f + \beta_p(R_m - r_f)]$$

Risk splits into two parts, and only the first is rewarded:

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_m^2}_{\text{systematic}} + \underbrace{\sigma_\varepsilon^2}_{\text{idiosyncratic}}$$

Annualising daily data: return $\times 252$, volatility $\times \sqrt{252}$.

Worked example. $r_f = 4\%$, $\beta = 1.2$, market risk premium $= 5.5\%$:

$$E[R] = 4\% + 1.2 \times 5.5\% = 10.6\%$$

This 10.6% is the **cost of equity** used in the WACC (section 9).

Real-world use

- **Cost of equity** for valuation, M&A and regulatory rate cases.
- **Fund evaluation.** A fund returning 12% with beta 1.5 when CAPM predicts 11% has alpha of +1%. A manager returning 12% with beta 0.5 has alpha of about +7%.

- **Risk reports.** Banks report the beta of their equity book to understand how much of the portfolio simply rides the market.

Interview questions. What does $\beta = 0$ mean? Can beta be negative? What are CAPM's weaknesses? (Single factor, one period, unobservable market; Fama-French adds size and value.) Sharpe versus Treynor? (Total risk versus market risk.)

6. Portfolio Theory (Markowitz)

Idea. Assets that do not move together reduce portfolio risk without reducing expected return. This is the "free lunch" of diversification.

Formulas

Two assets with weights w_1, w_2 and correlation ρ:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\,w_1 w_2\,\rho\,\sigma_1\sigma_2$$

N assets (vector of weights w, covariance matrix Σ, expected returns μ):

$$\mu_p = w^\top \mu \qquad\qquad \sigma_p^2 = w^\top \Sigma w$$

Efficient frontier (for each target return, find the lowest-risk portfolio), and the tangency portfolio:

$$\min_w \; w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu_{target}, \quad \sum w_i = 1 \qquad\qquad \max_w \; \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}}$$

Worked example. A 50/50 mix of A ($\sigma = 20\%$) and B ($\sigma = 30\%$) with $\rho = 0.3$:

$$\sigma_p^2 = 0.25(0.04) + 0.25(0.09) + 2(0.25)(0.3)(0.2)(0.3) = 0.0415 \;\Rightarrow\; \sigma_p = 20.4\%$$

That is lower than the 25% simple average. With $\rho = 1$ there is no benefit; with $\rho = -1$ risk can be eliminated.

Real-world use

- **Wealth management.** Advisors build a client portfolio on the efficient frontier for the client's risk tolerance: conservative clients get more bonds, aggressive clients more equity.
- **Bank loan books.** A bank lending to many industries is diversified. One concentrated in oil and gas is not: a price crash hits all borrowers together (high default correlation). Concentration limits come from this idea.
- **Robo-advisors** automate this optimisation.

Interview questions. Why do optimisers give extreme weights? (They exploit noise in estimated returns; fixes are constraints, shrinkage, Black-Litterman.) How does correlation behave in a crisis? (It rises, so diversification fails when needed most.)

7. Value at Risk (VaR) and Expected Shortfall (ES)

Idea. VaR answers: "What is the most I lose on 99% of days?" ES answers: "When I do have a bad day, how bad is it on average?"

Formulas

Parametric (normal) VaR for portfolio value V, mean μ, volatility σ per day, and confidence α:

$$VaR_\alpha = V\,(z_\alpha\,\sigma - \mu), \qquad z_{0.95} = 1.645,\ z_{0.99} = 2.326$$

Expected shortfall (normal case) and time scaling:

$$ES_\alpha = V(\sigma\,\frac{\varphi(z_\alpha)}{1-\alpha} - \mu) \qquad VaR_{T\text{ days}} = VaR_{1\text{ day}} \cdot \sqrt{T}$$

Historical VaR is simply the $(1-\alpha)$ percentile of past profit and loss. Monte Carlo VaR takes that percentile from simulated outcomes.

Worked example. A \$10M portfolio with daily volatility 1.5%:

$$VaR_{99\%,\,1\text{d}} = 10\text{M} \times 2.326 \times 0.015 = \$348{,}945 \qquad VaR_{99\%,\,10\text{d}} = 348{,}945 \times \sqrt{10} = \$1.10\text{M}$$

Backtest: at 99% you expect losses to exceed VaR on about 1% of days (roughly 2 to 3 days a year).

Real-world use

- **Daily risk limits.** A trading desk is told "99% 1-day VaR must stay under \$5M." The risk team reports it every morning.
- **Regulatory capital.** Basel rules link market-risk capital to VaR (now ES), and a bank with too many backtest exceptions has to hold more capital (the "traffic-light" test).
- **Why ES.** VaR ignores how large losses are beyond the threshold, and it can penalise diversification. ES is the Basel FRTB standard.

Method	Strength	Weakness
Parametric	Fast, closed form	Assumes normal returns, understates fat tails
Historical	No distribution assumption	Depends on the chosen window
Monte Carlo	Handles options and non-linear risk	Model-dependent, slower

Interview questions. Explain VaR to a non-technical manager. What are its limits? How do you backtest it? How does VaR differ from stress testing? (Stress testing applies a specific extreme scenario.)

8. Credit Risk: PD, LGD, EAD and Expected Loss

Idea. The core of lending. A bank expects some borrowers to default, and must price loans and set aside money for that.

Formulas

$$EL = PD \times LGD \times EAD$$

where PD is the probability of default, $LGD = 1 - \text{recovery rate}$, and
$EAD = \text{drawn balance} + CCF \times \text{undrawn limit}$.

Logistic regression model for PD (log-odds are linear in the borrower features):

$$\ln \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k \qquad \Leftrightarrow \qquad p = \frac{1}{1 + e^{-(\beta_0 + \beta^\top x)}}$$

A coefficient β_j is the change in log-odds per unit of x_j; e^{β_j} is the odds multiplier.

Model quality:

$$\text{Gini} = 2\,AUC - 1 \qquad KS = \max_s |F_{bad}(s) - F_{good}(s)|$$

Loans are priced so the rate covers funding, operating cost, expected loss and profit:

$$\text{min. rate} = \text{funding} + \text{opex} + PD \times LGD + \text{target margin}$$

Worked example. $PD = 2\%$, $LGD = 40\%$, $EAD = \$100{,}000$:

$$EL = 0.02 \times 0.40 \times 100{,}000 = \$800 \quad (0.8\% \text{ of the loan})$$

Pricing (funding 3.5%, opex 1.0%, margin 1.0%): a 2% PD loan needs $3.5 + 1.0 + 0.8 + 1.0 = 6.3\%$. A 19.8% PD loan needs about **13.4%**. In the notebook, the scorecard reaches **AUC 0.83** on held-out data.

Real-world use

- **Credit cards and personal loans.** Lenders score applicants (FICO, income, debt-to-income) and approve, price or decline automatically using a PD model.
- **Loan-loss provisions.** Under **IFRS 9** and **CECL**, banks book expected loss as an allowance. IFRS 9 stages: **Stage 1** performing (12-month ECL), **Stage 2** significant increase in credit risk (lifetime ECL), **Stage 3** credit-impaired (lifetime ECL). In the 2020 COVID shock, banks raised provisions sharply because forecast PDs rose.
- **Capital.** Expected loss is covered by pricing and provisions; unexpected loss (the volatility around EL) is covered by capital. Basel IRB capital is computed from PD, LGD and EAD.

Model validation checklist

- **Discrimination:** AUC, Gini, KS (does it rank risky borrowers above safe ones?).
- **Calibration:** predicted PD close to actual default rate by risk decile.
- **Stability:** Population Stability Index to detect drift after launch.
- **Imbalance:** with a 6.9% default rate, a model predicting "nobody defaults" is 93% accurate but useless, so use AUC or precision-recall, not accuracy.

Interview questions. Walk through building a PD model (target definition, features, split, validation, monitoring). EL versus UL? Through-the-cycle versus point-in-time PD? What counts as default under Basel? (90 days past due or unlikely to pay.)

9. Cost of Capital and DCF Valuation

Idea. A company is worth the present value of the free cash flow it will generate for all its capital providers, discounted at its cost of capital.

Formulas

$$WACC = \frac{E}{V} r_e + \frac{D}{V} r_d \,(1 - t) \qquad\qquad r_e = r_f + \beta \cdot MRP$$

$$FCFF = EBIT(1 - t) + D\&A - CapEx - \Delta NWC$$

$$EV = \sum_{t=1}^{N} \frac{FCFF_t}{(1 + WACC)^t} + \frac{TV}{(1 + WACC)^N} \qquad\qquad TV = \frac{FCFF_N \,(1 + g)}{WACC - g}$$

$$\text{Equity value} = EV - \text{Debt} + \text{Cash} \qquad\qquad \text{Value per share} = \frac{\text{Equity value}}{\text{Shares}}$$

Unlevering beta (to compare firms with different debt):

$$\beta_U = \frac{\beta_L}{1 + (1 - t)\, D/E}$$

Worked example (hypothetical company in the notebook). $r_e = 10.6\%$, $E = 3{,}600$, $D = 600$, $r_d = 6\%$, $t = 25\%$:

$$WACC = 0.857 \times 10.6\% + 0.143 \times 6\% \times 0.75 = 9.73\%$$

Year-5 free cash flow is \$356.1M and $g = 2.5\%$:

$$TV = \frac{356.1 \times 1.025}{0.0973 - 0.025} = \$5{,}048\text{M} \qquad PV(TV) = \frac{5{,}048}{1.0973^5} = \$3{,}173\text{M (73\% of } EV)$$

$EV = \$4{,}338$M, equity $= 4{,}338 - 600 + 150 = \$3{,}888$M, so **\$38.88 per share** versus a \$36 market price.

Real-world use

- **Mergers and acquisitions.** An acquirer values the target with a DCF, then compares it with the offer price. Bankers show a WACC versus growth grid (here, 1 percentage point lower WACC lifts value from \$38.9 to \$46.0).
- **Equity research and investing.** Analysts publish target prices from DCFs; investors compare to market price.
- **Credit analysis.** Lenders use FCFF forecasts to test whether a borrower can service debt.
- **Banks are different.** A bank's debt is raw material, not financing, so FCFF-DCF does not fit. Analysts value banks with a **dividend discount** or **residual income** model, using the justified price-to-book ratio:

$$\frac{P}{B} = \frac{ROE - g}{r_e - g}$$

Interview questions. Walk me through a DCF in two minutes. Why does terminal value dominate? How does more debt affect WACC? Sanity checks: growth below long-run GDP, WACC above g, implied exit multiple reasonable.

10. Monte Carlo Simulation

Idea. Replace single-number assumptions with probability distributions, run the model thousands of times, and read the answer as a distribution.

Formulas

Simulate an asset price (geometric Brownian motion), where $Z \sim N(0, 1)$:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right]$$

Accuracy of an estimate from N runs (standard error shrinks slowly):

$$SE = \frac{s}{\sqrt{N}}$$

Worked example. The notebook runs 10,000 DCFs, drawing growth, margin, beta and terminal growth at random. Result: median **$38.8 per share**, 5th–95th percentile **$26–$56**, and a **62.9% probability** that value exceeds the $36 market price.

Real-world use

- **Stress testing and capital planning.** Banks simulate thousands of interest-rate, unemployment and house-price paths to see the loss distribution for regulators.
- **Pricing complex derivatives** that have no closed-form formula.
- **Investment committees** prefer "63% chance the stock is undervalued" to a single point estimate.

Pitfalls. Inputs are often correlated, so sampling them independently understates risk. The answer is only as good as the chosen distributions.

11. Bank-Specific Concepts

Concept	Formula or definition	Real-world meaning
Net interest margin (NIM)	$\frac{\text{interest income} - \text{interest expense}}{\text{avg earning assets}}$	Core profitability of lending; rises when rates rise for asset-sensitive banks
Efficiency ratio	$\frac{\text{non-interest expense}}{\text{revenue}}$	Lower is better; around 55–60% is typical
DuPont ROE	$\frac{NI}{Rev} \times \frac{Rev}{Assets} \times \frac{Assets}{Equity}$	Shows how leverage drives bank returns
CET1 ratio	$\frac{\text{common equity tier 1}}{\text{risk-weighted assets}}$	Capital buffer; minimum 4.5% plus buffers (roughly 7%+)
Leverage ratio	$\frac{\text{Tier 1 capital}}{\text{total exposure}}$	Simple backstop to risk-weighted rules (minimum 3%)
LCR	$\frac{\text{high-quality liquid assets}}{\text{30-day net outflows}} \geq 100\%$	Can the bank survive a month of stress?
NSFR	$\frac{\text{available stable funding}}{\text{required stable funding}} \geq 100\%$	Is long-term lending funded by stable sources?
Funds transfer pricing	Internal rate charged for funding	Shows each product's true margin

DuPont example. Net income/revenue $= 25\%$, revenue/assets $= 4\%$, assets/equity $= 11$:

$$ROE = 0.25 \times 0.04 \times 11 = 11\%$$

High leverage drives bank ROE, which is why regulators set capital floors.

Interest-rate risk in the banking book (IRRBB). Banks borrow short (deposits) and lend long, creating a duration gap. Approximately:

$$\Delta EVE \approx -(D_A - \tfrac{L}{A} D_L) \cdot A \cdot \Delta y$$

where A and L are assets and liabilities. If asset duration exceeds liability duration, rising rates cut the economic value of equity. Banks measure this with repricing gaps, duration gaps and net-interest-income simulations under ± 100bp shocks.

Model risk. Regulators (SR 11-7 in the US) expect every model, such as your PD scorecard, to be independently validated and monitored. The **three lines of defence** are the business, risk and compliance oversight, and internal audit.

12. Quick-fire Interview Answers

1. **Single most important idea in finance?** Value equals the present value of future cash flows, adjusted for risk.
2. **Rates rise: what happens to a bank?** Bond values fall, deposit costs rise, prepayments slow, credit risk increases. Net interest income rises if the bank is asset-sensitive.
3. **Explain your project simply.** “I built Python models that answer how much a loan costs, how risky a portfolio is, what a company is worth, and how much to set aside for loans that go bad.”
4. **What would you change with real data?** Out-of-time validation, drift monitoring, macro-linked PDs, and documentation for model risk.
5. **Weakest part of your DCF?** Terminal value is about 73% of EV and very sensitive to WACC and g, hence the sensitivity grid and Monte Carlo.
6. **Why does correlation matter?** It sets how much diversification you get, and it spikes in a crisis, which is exactly when VaR can fail.

Two-week Study Plan

- **Days 1–3:** sections 1–4. Re-derive each formula by hand once.
- **Days 4–6:** sections 5–7. Re-run the notebook modules with different parameters.
- **Days 7–9:** sections 8 and 11, your differentiator given your bank job.
- **Days 10–12:** sections 9–10. Practise a two-minute DCF walk-through aloud.
- **Days 13–14:** mock interviews from section 12, and explain each notebook module in 60 seconds.